

ScoreVision

Bowling Scoreboard Data Extraction - Assignment Documentation

Objective

Build a Computer Vision pipeline that reads the bowling scoreboard from a video, performs OCR on multiple frames, and exports the extracted observations as structured JSON and CSV.

Input video analysis

The supplied recording is 1920 x 1080 at 30 FPS, approximately 57.83 seconds. The scoreboard stays in a stable screen position but is interrupted by bowling animations, so a calibrated ROI and frame sampling strategy are used.

6	TARUN										
	1	2	3	4	5	6	7	8	9	10	TTL
J	X	5 -	- 7	4 -							0
	15	20	27	31							31
V	8 -	3 -	7 1	8 1							0
	8	11	19	28							28
P	X	4 /	9 -	6 -							0
	20	39	48	54							54
T	6 1	1 /	8 -								0
	7	25	33								33
2.5											

Figure 1. Representative input frame from the supplied video.

Scoreboard detection

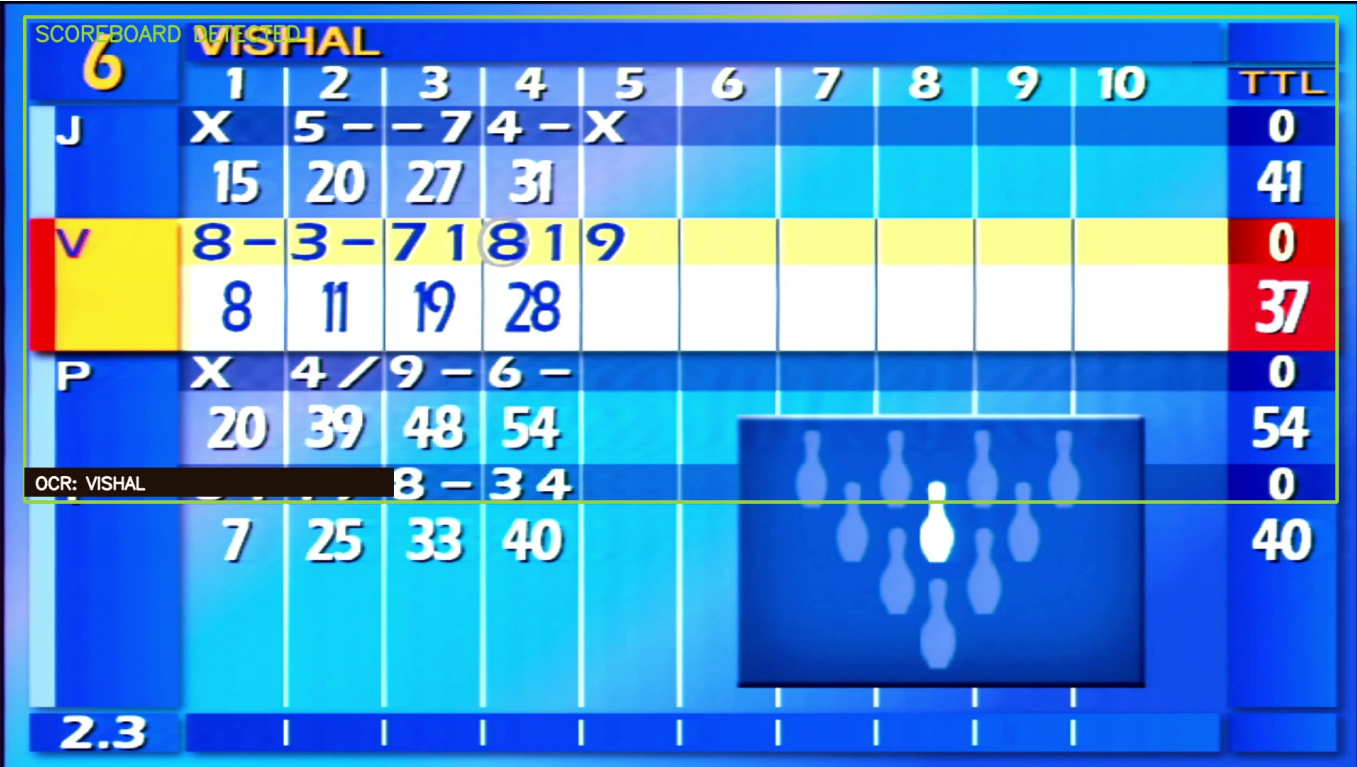


Figure 2. Calibrated scoreboard ROI shown by the green bounding box.

Pipeline

OpenCV video reader -> frame sampling -> scoreboard presence check -> calibrated ROI crop -> preprocessing -> Tesseract OCR -> multi-frame observation selection -> JSON/CSV export.

OCR / extracted observations

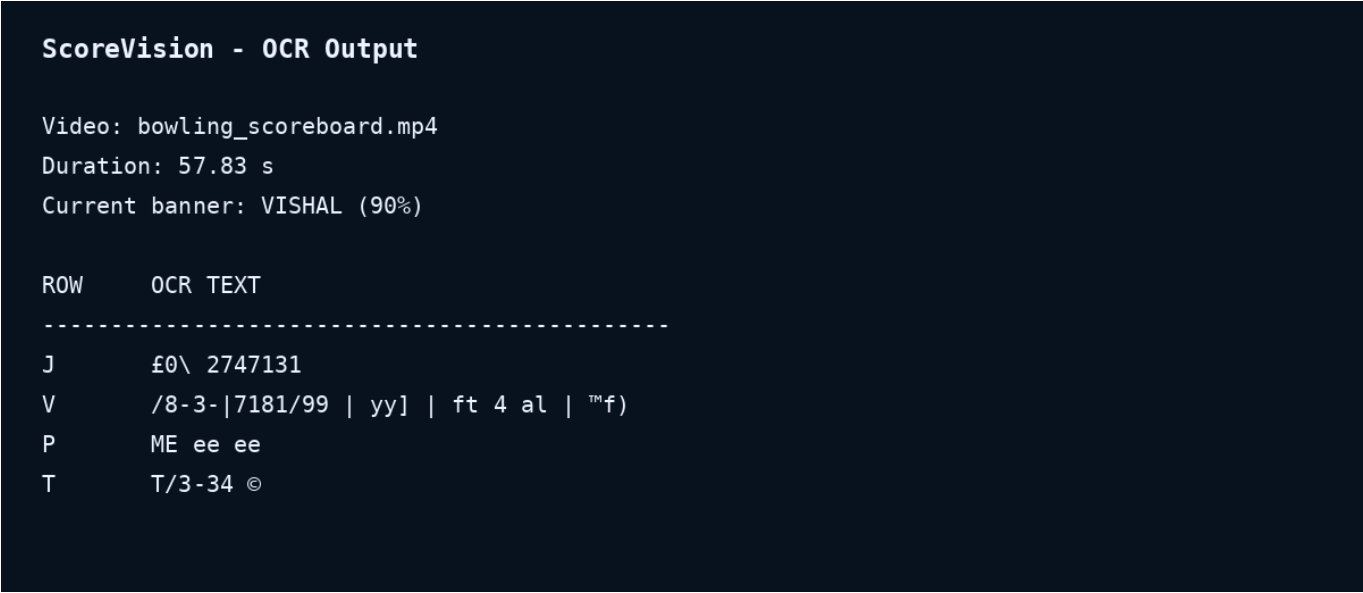


Figure 3. OCR observations returned from the selected scoreboard frames. The scoreboard font is stylized, so some raw OCR tokens can remain noisy; the output keeps confidence and raw text instead of inventing values.

Final output

Player	OCR text	Numbers detected	Confidence	Frames
J	£0\ 2747131	0	54%	8
V	/8-3-[7181/99 yy] ft 4 al ™f	8, 3, 99, 4	46%	8
P	ME ee ee	-	40%	8
T	T/3-34 ©	3, 34	61%	8

Artifacts generated: scoreboard_data.json, scoreboard_data.csv, annotated/best_frame.jpg and annotated/scoreboard_crop.jpg.

Run locally with: python app.py, then open <http://127.0.0.1:5000>.